

MAT_SCI 465: Advanced Electron Microscopy & Diffraction

Week 05 & 06 Combined Assignment

HRTEM Simulation and Contrast Transfer Function (CTF) Analysis

Assignment notebook: https://github.com/NU-MSE-LECTURES/465-WINTER2026/blob/main/Week_06/assignments/assignment_combined_5_6.ipynb

Deadline: 03/20/2026

Software & Package

abTEM (ab initio Transmission Electron Microscopy) - a GPU-accelerated Python package for simulating TEM and STEM images using the multislice algorithm.

Package documentation: <https://abtem.readthedocs.io/>

Objective

Use the **abTEM** package to perform a multislice High-Resolution TEM (HRTEM) simulation on a Silicon Nitride (Si_3N_4) crystal. You will build the atomic model, compute the projected electrostatic potential, propagate the electron wave through the specimen, and apply Contrast Transfer Functions (CTFs) under different microscope conditions. The final task includes a study of Poisson noise and its effect on image interpretability. The written answers to the four self-reflection questions are NOT MANDATORY!!

Task 1 · Projected Potential Calculation

- ☐ Create a **FrozenPhonons** object using the orthogonalized Si_3N_4 unit cell tiled by $(3 \times 2 \times 17)$, giving a specimen thickness of ~ 5 nm. Use 8 configurations and a thermal displacement parameter $\sigma = 0.1$ Å for both Si and N atoms.
- ☐ Create a **Potential** object with a real-space sampling of 0.1 Å, a slice thickness of 2.0 Å, and the 'kirkland' parametrization.
- ☐ Plot the mean projected potential using a 'viridis' colormap.

The provided notebook includes:

- Code to build the Si_3N_4 unit cell via `ase.Atoms` and `abtem.structures.orthogonalize_cell`.
- Visualization of the tiled structure (`abtem.show_atoms`) in the xy -plane.
- Starter code with # TODO markers for **FrozenPhonons**, **Potential**, and the mean-projected potential plot.

Task 2 · Multislice Exit-Wave Simulation

- ☐ Initialize a **PlaneWave** object with an accelerating voltage of 300 keV.
- ☐ Run the multislice algorithm using the **Potential** computed in Task 1 to obtain the complex exit waves.
- ☐ Compute the mean intensity across phonon configurations (`exit.waves.intensity().mean(0)`) and visualize with `cmap='viridis'`.

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The notebook includes:

- Starter code with # TODO markers for PlaneWave initialization, multislice propagation, and mean-intensity plotting.

Task 3 · Contrast Transfer Function (CTF)

- ☐ Define an **uncorrected CTF** with defocus $C_{10} = -600 \text{ \AA}$ and spherical aberration $C_{30} = 1.3 \times 10^7 \text{ \AA}$ ($C_s = 1.3 \text{ mm}$).
- ☐ Define an **aberration-corrected CTF** with $C_{10} = 30 \text{ \AA}$ and $C_{30} = -8 \times 10^4 \text{ \AA}$.
- ☐ Plot both CTFs as a function of spatial frequency and discuss the difference in information transfer (contrast envelope, zero crossings).
- ☐ Apply each CTF to the exit waves via `exit_waves.apply_ctf(ctf)` and compare the resulting mean image intensities side by side.

The notebook includes:

- Template code for `abtem.CTF(energy=300e3, parameters={...})` for both uncorrected and corrected cases.
- # TODO stubs for CTF definition, combined CTF plot, application to exit waves, and resulting image visualization.

Task 4 · Noise and Electron Dose

- ☐ Apply Poisson noise to the aberration-corrected image using `abtem.noise.poisson_noise` with a dose of $100 \text{ e}^-/\text{\AA}^2$.
- ☐ Display the noiseless and noisy images side by side.
- ☐ Comment on how the signal-to-noise ratio changes and how this affects the visibility of atomic columns.

The notebook includes:

- Import statement from `abtem.noise import poisson_noise` and # TODO stubs for applying the dose and producing the comparison figure.

Task 5 (Self-Reflection) · Conceptual Questions

Answer the following questions *within your notebook* as Markdown cells (no separate submission required, but responses will be reviewed):

1. **CTF comparison:** How does the uncorrected CTF differ from the aberration-corrected CTF in terms of spatial-frequency transfer? Why do atomic columns appear sharper in the aberration-corrected image?
2. **Defocus and contrast:** Why is imaging at exactly zero defocus ($\Delta f = 0$) typically avoided in conventional (uncorrected) TEM? What is the significance of the Scherzer defocus?

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- Noise and interpretability:** How does adding Poisson noise change the interpretability of the image? Which other experimental factors (not simulated here) would further degrade a real HRTEM image?
 - Multislice vs. kinematic approximation:** Why is the multislice method necessary for a 5 nm-thick specimen, rather than simply using a single projected potential?

Dataset: Si₃N₄ Crystal Model

- Structure:** α -Si₃N₄ unit cell defined analytically via `ase.Atoms` with cell parameters [7.6045, 7.6045, 2.9052] Å and angles [90, 90, 120].
- The orthogonalized and tiled supercell ((3×2×17)) is constructed within the notebook; no external data download is required.
- abTEM version $\geq 1.0.0$ beta34 is required. Install via:

```
pip install abtem
```

Full documentation: <https://abtem.readthedocs.io/>

Key Parameters Reference

Parameter	Symbol	Uncorrected	Aberration-Corrected
Accelerating voltage	V		300 keV
Defocus	C_{10} (Δf)	-600 Å	$+30$ Å
Spherical aberration	C_{30} (C_s)	1.3×10^7 Å (1.3 mm)	-8×10^4 Å
Electron dose (noise)	-		$100 e^-/\text{Å}^2$
Slice thickness	-		2.0 Å
Real-space sampling	-		0.1 Å
Thermal σ (Si, N)	-		0.1 Å

Deliverables Checklist

- ☐ **Projected potential:** Plot of the mean projected potential of the Si₃N₄ supercell (Task 1).
- ☐ **Exit-wave intensity:** Mean intensity map from the multislice simulation (Task 2).
- ☐ **CTF comparison figure:** Side-by-side CTF plots (uncorrected vs. corrected) as a function of spatial frequency (Task 3).
- ☐ **HRTEM images:** Mean image intensity maps for both the uncorrected and the aberration-corrected CTF conditions (Task 3).
- ☐ **Noise comparison figure:** Noiseless vs. noisy ($100 e^-/\text{Å}^2$) aberration-corrected images (Task 4).
- ☐ Written answers to the four self-reflection questions embedded as Markdown cells in the notebook (Task 5).
- ☐ Publication-quality figures with labeled axes (Å or nm scale bars where applicable) and submission of the completed .ipynb file via Canvas.